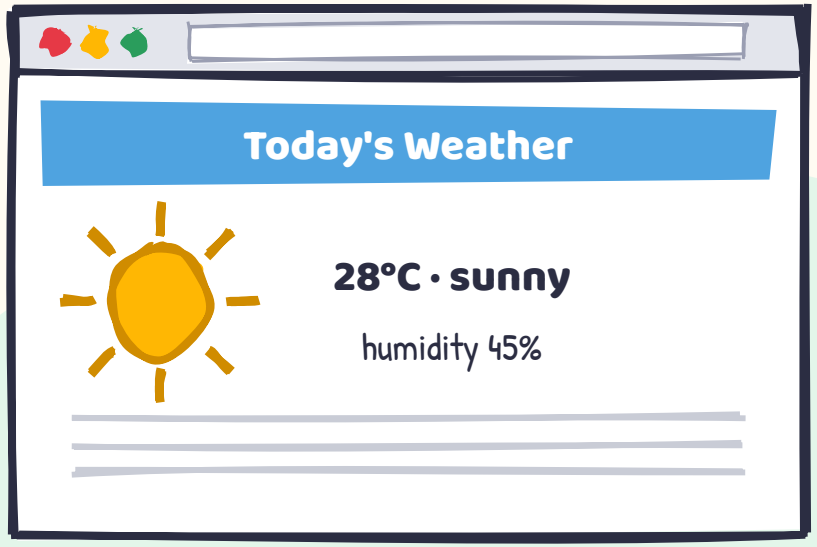
Junior Makers · Class 14

Weather Report Webpage

Build your very first website!

HTML structure | CSS style | Real weather data



Did you know?
Every website you visit, even YouTube, is built from HTML, the language we learn today.

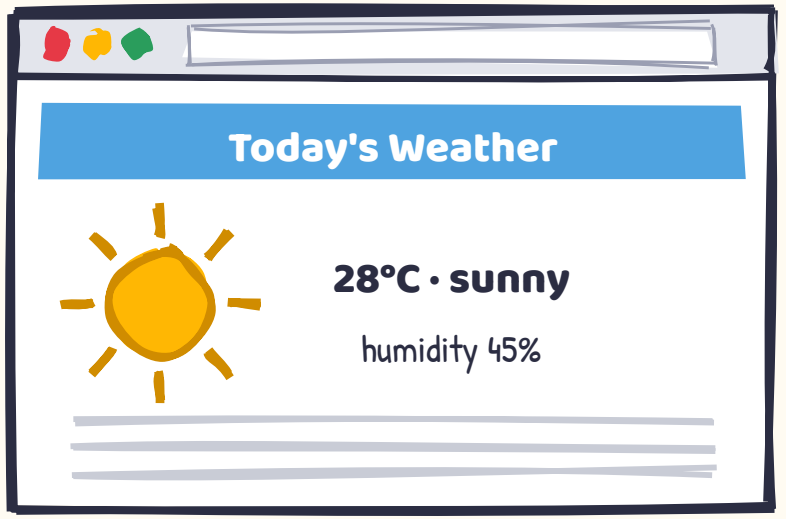
Let's think...

What is a website made of?

A secret

A webpage looks like pictures and colours, but underneath it is just **text**: special words that tell the browser what to show.

Teacher demo: right-click a page, choose **View page source**.



Pictures?

Magic code?

Just TEXT!

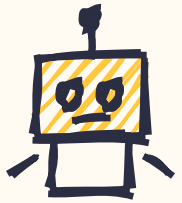
Today: we turn the readings from our **Mini Weather Station** (JM-03) into a real weather report webpage.

Learning aims

Today I will learn to...

1. Write **HTML tags** for headings and paragraphs
2. Add an **image** with a helpful description
3. Use **CSS** to change colours and fonts
4. Show **real weather readings** in a list
5. Present data clearly in a **table**

Keywords
HTML · tag · heading · paragraph · image · CSS · style



Watch and think

Video time!



The Internet: HTTP & HTML

Code.org · about 7 minutes · youtu.be/kBXQZMmiA4s

Watch for...

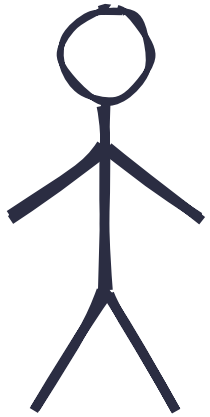
1. What is **HTML** used for?
2. What happens when you type a web address?
3. What is a **tag**?

Scan to watch

After the video: Write the tag you would use for a big heading on your weather page.

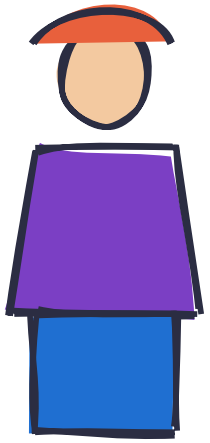
Big idea

HTML is the skeleton, CSS is the clothes



HTML

WHAT is on the page:
headings, text, pictures



CSS

HOW it looks:
colours, fonts, sizes

HTML means

Hyper**T**ext **M**arkup **L**anguage. **CSS** means **C**ascading **S**tyle **S**heets.

Coming later

JavaScript makes pages DO things. Young Innovators learn it!

Real world

Who makes webpages?



Weather sites
Show today's temperature and tomorrow's rain



School websites
News, timetables and events



Game sites
Rules, scores and news for players



Web designers
A real job: building pages people love to use



What is your favourite website? What makes it easy (or hard) to use?

Big number
There are over 1 billion websites in the world!

LEARN

Meet the tags

Tags are like a sandwich

```
<h1>Hello!</h1>
```

opening tag	content	closing tag
bread on top	the filling	bread below (with /)

Don't forget!
Every opening tag needs a closing tag with a slash /

Tag	What it makes
…	biggest heading
…	smaller heading
…	a paragraph of text
…	bold words

Almost every tag
has a closing partner. One exception today: the image tag.

LEARN

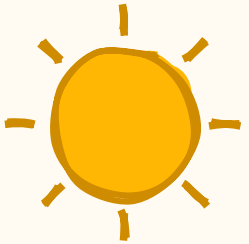
Add a picture

The image tag

```
<img src="sun.png"
     alt="A bright yellow sun"
     width="200">
```

Why alt text matters

People who are **blind** use screen readers that read pages out loud. The **alt** text tells them what the picture shows. It also appears if the picture fails to load.



Part	Meaning
src	where the picture is (its link or file name)
alt	words that DESCRIBE the picture
width	how wide, in pixels

No closing tag!

The image tag is EMPTY: nothing goes inside, so no </img>.

Style it

CSS: give your page clothes

```
body {
  background-color: lightblue;
  font-family: Arial;
}
h1 {
  color: navy;
  text-align: center;
}
p {
  font-size: 20px;
}
```

How to read CSS

h1 = WHO to style

color = WHAT to change

navy = the new VALUE

Every line ends with a **;**

American spelling!

CSS writes color and center, not colour and centre.

Try these colours

tomato · gold · seagreen · orchid · skyblue · or codes like **#FFB703**

Where?

Type CSS in the CSS box of the editor, not the HTML box.

LEARN

Level 2

Compare readings in a table

```
<table border="1">
  <tr><th>Place</th><th>Temp</th></tr>
  <tr><td>Classroom</td><td>27 °C</td></tr>
  <tr><td>Playground</td><td>31 °C</td></tr>
</table>
```

Why a table?

Tables make it easy to **compare**. Which place was hotter? You can see it in one second!

Place	Temp
Classroom	27 °C
Playground	31 °C

Table tags

tr = table row
th = heading cell
td = data cell

Add a row

Measure one more place and add a new <tr> row.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

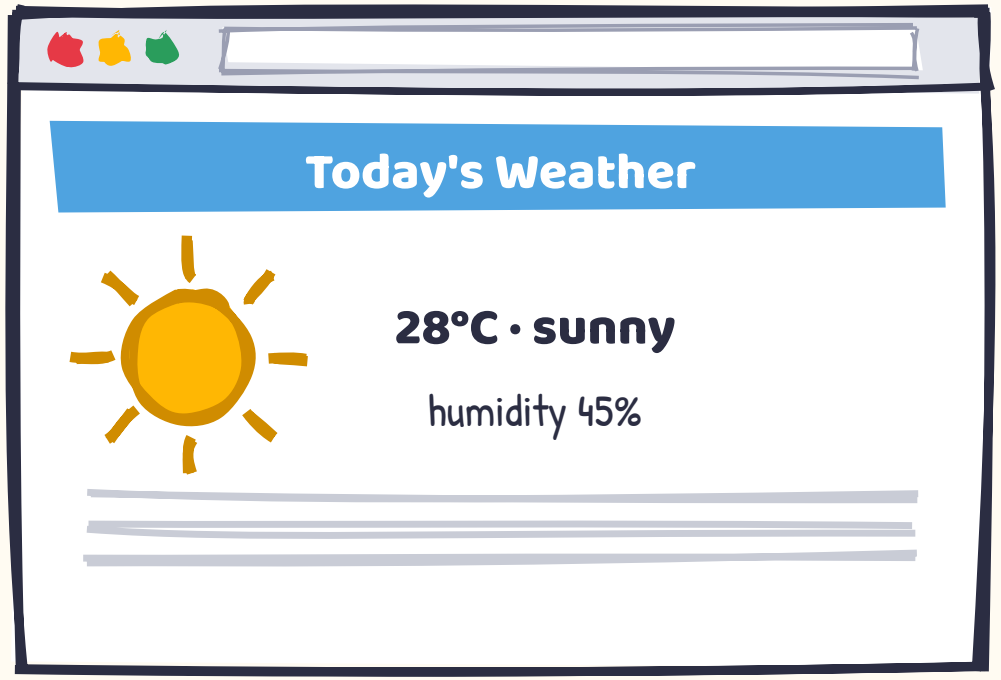
1. Your first webpage
2. Human HTML!
3. Add your weather station readings
4. Make it easy to read
5. Bug hunt!
6. Web designer challenges

MAKE

Code it

Your first webpage

```
<h1>Weather Report</h1>
<h2>Pak-Turk Maarif, Islamabad</h2>
<p>Today is <b>sunny</b> and warm.</p>
<p>Reported by: Team Falcon</p>
```



Where to type

Open **CodePen** (or the online HTML editor your teacher shows). Type in the **HTML** box. The page appears **live** as you type!

Safety: use a team name, not your full name, on pages that might go online.

Class game

Human HTML!

How to play

1. Cards are handed out: **<h1>**, **</h1>**, ** **, ** ** and word cards.
2. Teams race to stand in the right **order** to build a heading and a paragraph.
3. The "browser" (teacher) reads it out. A mistake? The page breaks!
4. Fastest correct team wins.

Then discuss

What happened when someone forgot the **closing** tag?



Big idea

Browsers follow instructions EXACTLY. One missing / can change the whole page.

MAKE

Real data

Add your weather station readings

```
<h2>Today's readings</h2>
<ul>
  <li>Temperature: 28 °C</li>
  <li>Humidity: 45 %</li>
  <li>Light level: 812</li>
</ul>
```

List tags

<ul> = a bullet list (unordered)

<li> = one list item

<ol> = a numbered list

Be honest

Real reporters only publish real measurements!

Where do the numbers come from?

Use the readings your team wrote down from the **Mini Weather Station** (JM-03). No readings? Your teacher will share a class set.

Units!

28 what? Always write °C or % so readers know what the number means.

MAKE

Design like a pro

Make it easy to read

- **Contrast:** dark text on a light background (or light on dark).
- **Few colours:** pick 2 or 3 colours that go well together.
- **Big enough:** text at least 16 px so everyone can read it.
- **Headings first:** readers scan headings to find what they need.
- **Alt text:** every picture gets a description.



Test it

Ask a friend to find today's temperature in 5 seconds. Can they?

Fix it

Bug hunt!

What you see	Likely bug
Whole page turned into a heading	missing </h1>
Picture shows a broken icon	wrong src name or link
Colour does not change	spelled colour, or missing ;
Style does nothing at all	CSS typed in the HTML box
Text runs together	forgot <p> tags

Bug hunter tip

Change **one thing** at a time, then look. Then you know exactly what fixed it.

Level up!

Web designer challenges

Choose your challenge

- **Forecast:** add a section for tomorrow with your own prediction.
- **Theme:** make a hot-day red/orange style and a cool-day blue style.
- **Emoji weather:** use symbols like ☀ ☁ ☂ next to each reading.
- **Link it:** add a link to a real weather website with <a href="...">.

Think like an engineer

A real weather website updates every hour by itself. How could our weather station send numbers to a page automatically?

Boss level

Answer: an ESP32 with Wi-Fi can host a live page. You will build one in Young Innovators!

Show what you know

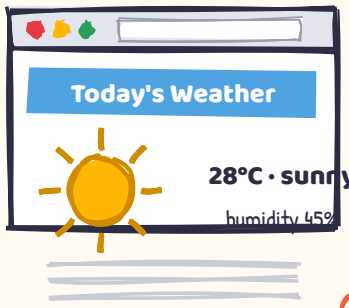
What did we learn?

- [x] HTML says WHAT is on the page, CSS says HOW it looks
- [x] Tags come in pairs: <h1> and </h1>
- [x] Images need src and helpful alt text
- [x] Lists and tables make data easy to read
- [x] Always write units like °C and %

Exit ticket: what is the difference between HTML and CSS?

At home

Look at your favourite website. Find one heading, one picture and one list.



Next class: Pong Game with a Knob Controller!